EQUITY
vs EQUALITY

Two ideas of fairness that quietly pull apart

A VADEMECUM



Same treatment, or same outcome? You often cannot have both.

The core distinction

EQUALITY

Treat everyone the same

Same rules, same resources, same process — applied identically to all.

- About sameness of INPUT.
- Simple, clear, hard to abuse — anyone can check it.
- But ignores that people start from different places.

EQUITY

Give each what they need

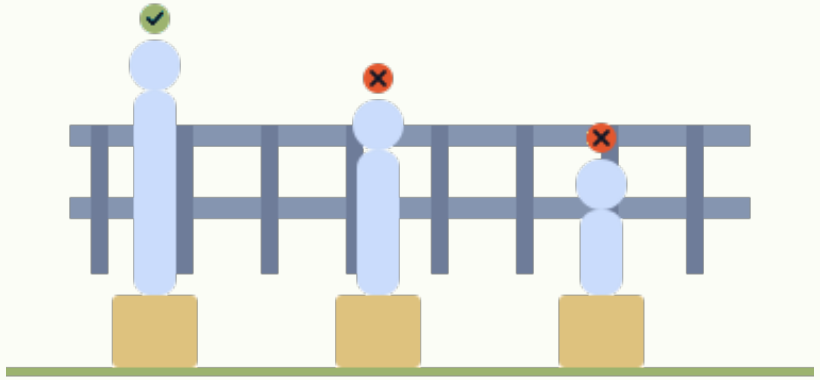
Different resources, calibrated to different starting points.

- About fairness of OUTCOME.
- More just in principle — accounts for real disadvantage.
- But someone must decide who needs more, and why.

The fence

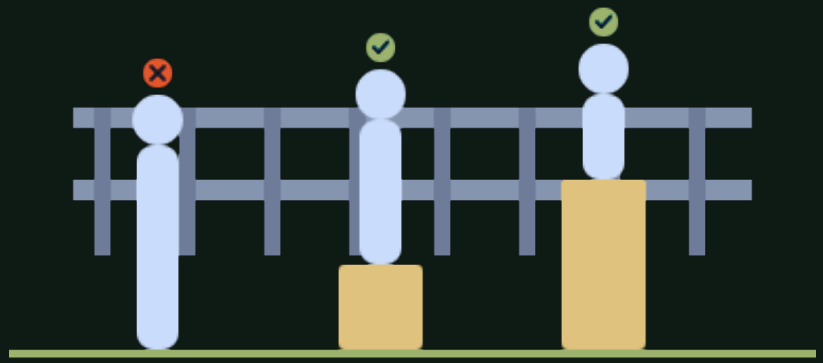
Three people of different heights, one fence. Same goal — everyone sees over. Opposite method.

EQUALITY — same boxes



Identical help for all — the tallest is fine, the shortest still can't see.

EQUITY — calibrated boxes



Different help each — none for the tallest, most for the shortest. All see.

Why they collide

The fence proves something uncomfortable: you cannot maximise both at once.

≠ Want equal OUTCOMES?

Everyone sees over the fence — then you must give UNEQUAL resources (different boxes).

= Want equal RESOURCES?

Everyone gets the same box — then you must accept UNEQUAL outcomes (some see, some don't).

This is the same truth the fairness impossibility theorem states in mathematics — the metaphor and the theorem are one fact at two levels of precision.

A real collision: COMPAS

A US court risk-scoring tool. Two sides each called it (un)fair — and both were right.

ProPublica said: unfair

- More false positives for Black defendants — wrongly flagged high-risk.
- Measures: equal ERROR RATES across groups.

Northpointe said: fair

- A given score meant the same reoffend-probability for everyone.
- Measures: equal CALIBRATION across groups.

When base rates differ between groups, you mathematically cannot have both. Neither side lied — they measured different fairnesses.

Sources: ProPublica, 'Machine Bias' (2016); Chouldechova (2017); Kleinberg, Mullainathan & Raghavan (2016).

The hard part

Equity is more just — but it hands someone the power to decide who deserves more.



Which gaps count?

Height is obviously relevant and blameless. Real disadvantages are contested — deciding which to correct is a moral, not a technical, claim.



Who holds the power?

Equity requires an authority to define 'need.' In the wrong hands, 'equity' can justify almost any differential treatment.



AI forces the choice open

You must ENCODE which fairness you mean in the model. AI can't be vague — it drags a choice societies leave blurry into the open.

The non-discrimination paradox

The two ideas give opposite instructions — both flying the flag of 'fairness.'

Equality says:

- Be BLIND to the protected trait — don't look at race or gender.
- Fails: the proxy problem. The model rebuilds the trait from ZIP code, name, history.

Equity says:

- SEE the protected trait — and actively correct for its disadvantage.
- Fairer outcomes, but treats groups differently — which can itself be called discrimination.

"Don't look at race" vs "you must look at race to undo its effects." That is the whole fight.

Vademecum — the essentials

Equality

Same treatment for all. About equal INPUT. Simple and hard to abuse.

Equity

What each needs to reach the same place. About equal OUTCOME.

They conflict

Same boxes or same view — you can't fully have both. It's a forced choice.

The choice is political

Deciding which disadvantage to correct is a moral claim, not a technical one.

Blindness fails

Ignoring the trait doesn't remove it — proxies rebuild it. Non-discrimination may require seeing it.

AI makes it explicit

A model forces you to encode which fairness you mean — no more hiding the choice.

"Make it fair" is never one instruction — it is a choice about which fairness, and who it's for.